

Fortran Project - Week 2

IF, DO, and Arrays - Store and Analyze Repeated Data

Today's goal

Extend Week 1 so that the program can process an arbitrary number of measurements with loops and conditions, then store the measurements in an array so that the same data can be reused for further analysis.

1. Guided Coding

Study the example code below with your team. Type it into your editor and run it. Do not copy and paste.

[Part A] - Process repeated input with DO

```
program week2_guided_a
  implicit none

  integer :: i, n
  double precision :: x, sum, mean

  print *, 'Number of measurements:'
  read *, n

  sum = 0.0d0

  do i = 1, n
    print *, 'Enter measurement ', i
    read *, x
    sum = sum + x
  end do

  mean = sum / dble(n)

  print *, 'Mean = ', mean
end program week2_guided_a
```

[Part B] - Store the same input in an array

In Part A, each value is used immediately and then replaced by the next value.

Modify the program so that every measurement is stored and can be analyzed again later.

```
program week2_guided_b
  implicit none

  integer, parameter :: max_data = 1000
  integer :: i, n
  double precision :: data(max_data)
  double precision :: sum, mean

  print *, 'Number of measurements:'
  read *, n

  do i = 1, n
    print *, 'Enter measurement ', i
    read *, data(i)
  end do

  sum = 0.0d0

  do i = 1, n
    sum = sum + data(i)
  end do
```

```
mean = sum / dble(n)

print *, 'Mean = ', mean
end program week2_guided_b
```

2. Team Coding Tasks

1. Start from Part B and keep every measurement in data(i).
2. Calculate maximum and minimum by looping over the array.
Initialize both values safely using the first measurement.
3. Calculate the population standard deviation: $\sqrt{\text{sum}((\text{data}(i) - \text{mean})^2) / \text{dble}(n))}$.
4. Ask the user for a threshold.
5. Loop over the stored data and print the index and value of every measurement greater than the threshold.
6. Also count the total number of measurements greater than the threshold.
7. Run the completed program from beginning to end and keep it executable at the end of the session.

3. Expected Output / Checkpoint

```
Number of measurements:
5

Enter measurement 1
12
Enter measurement 2
17
Enter measurement 3
8
Enter measurement 4
21
Enter measurement 5
14

Enter threshold:
15

Index   Value
-----
2        17.0
4        21.0

Mean = 14.4
Std. Dev. = ...
Maximum = 21.0
Minimum = 8.0
Above threshold = 2
```

4. Submission Check

- The program compiles without errors.
- The program runs from beginning to end.
- Every measurement is stored in the array and can be reused.
- Every team member can explain the DO loops, IF statements, and array indexing used this week.
- Save the completed source code as week2_project.f90.